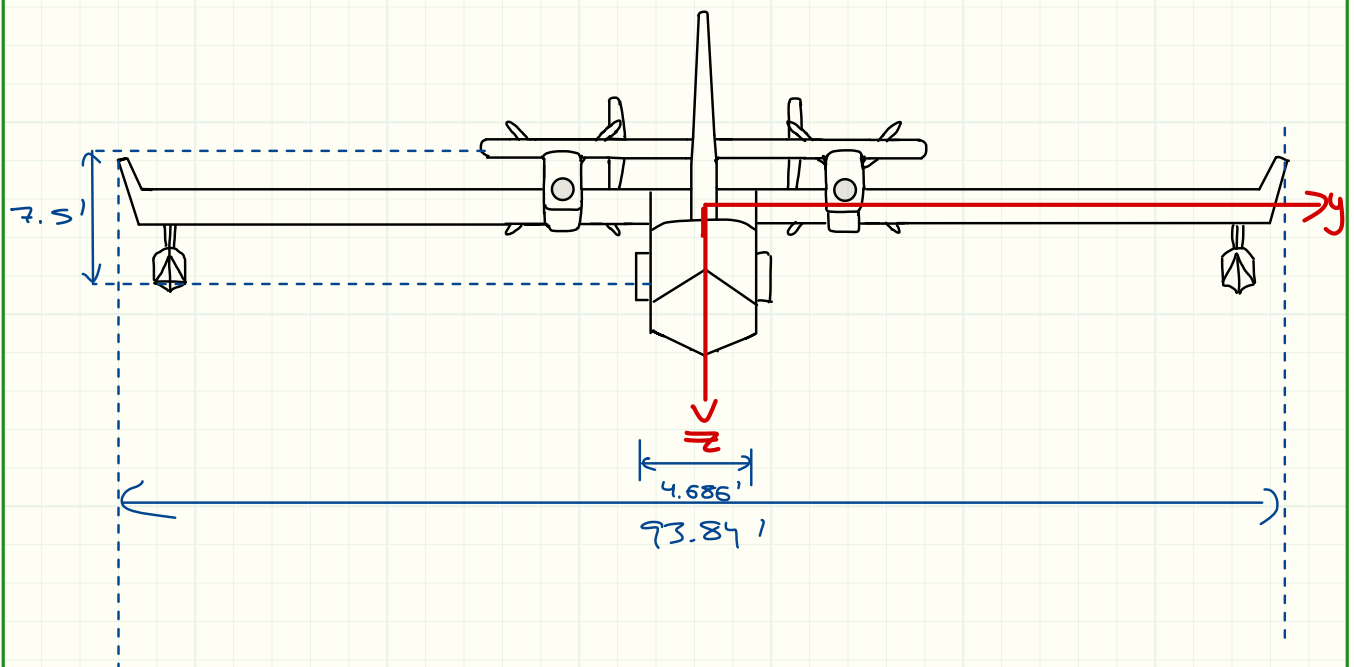
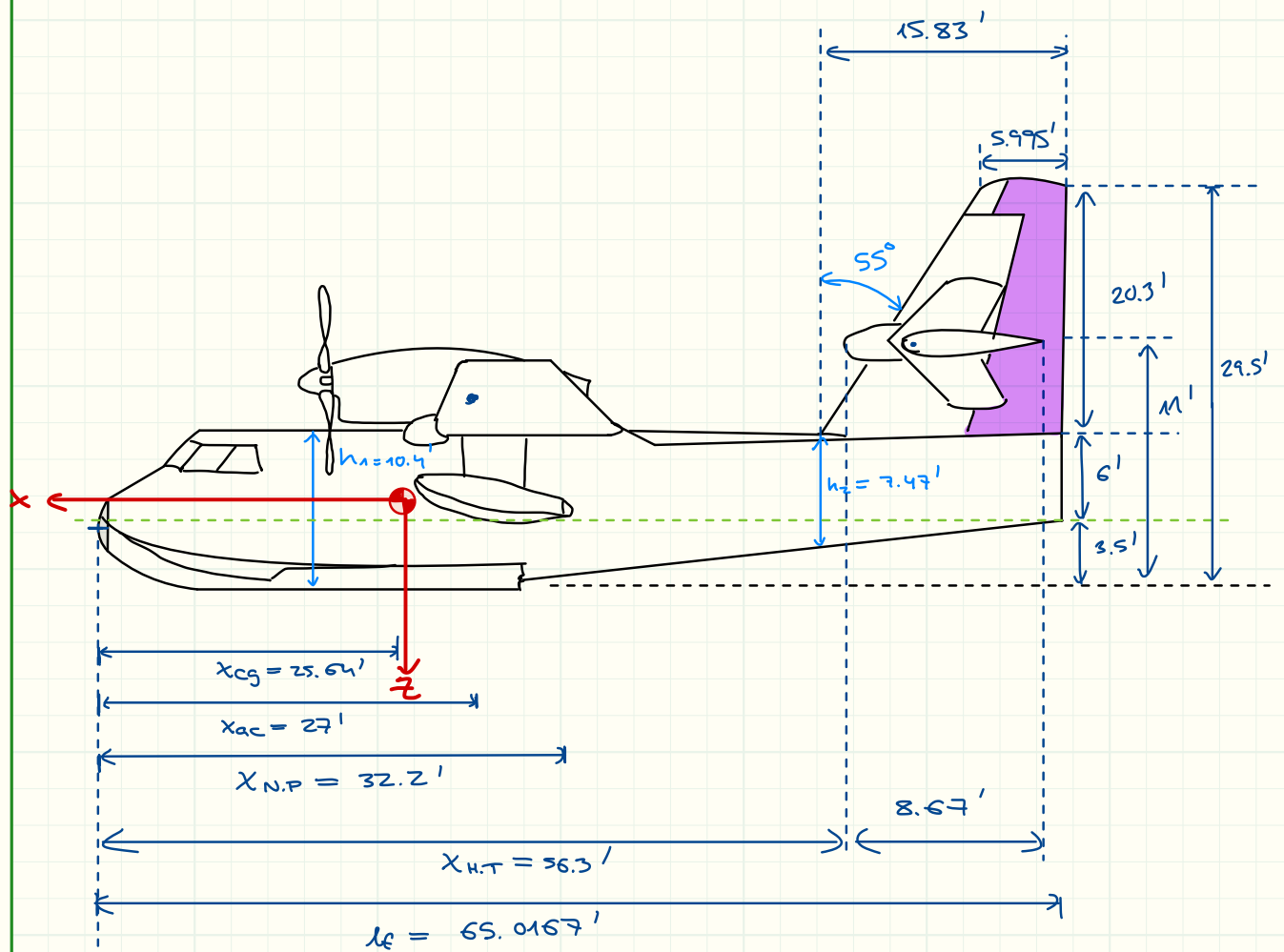
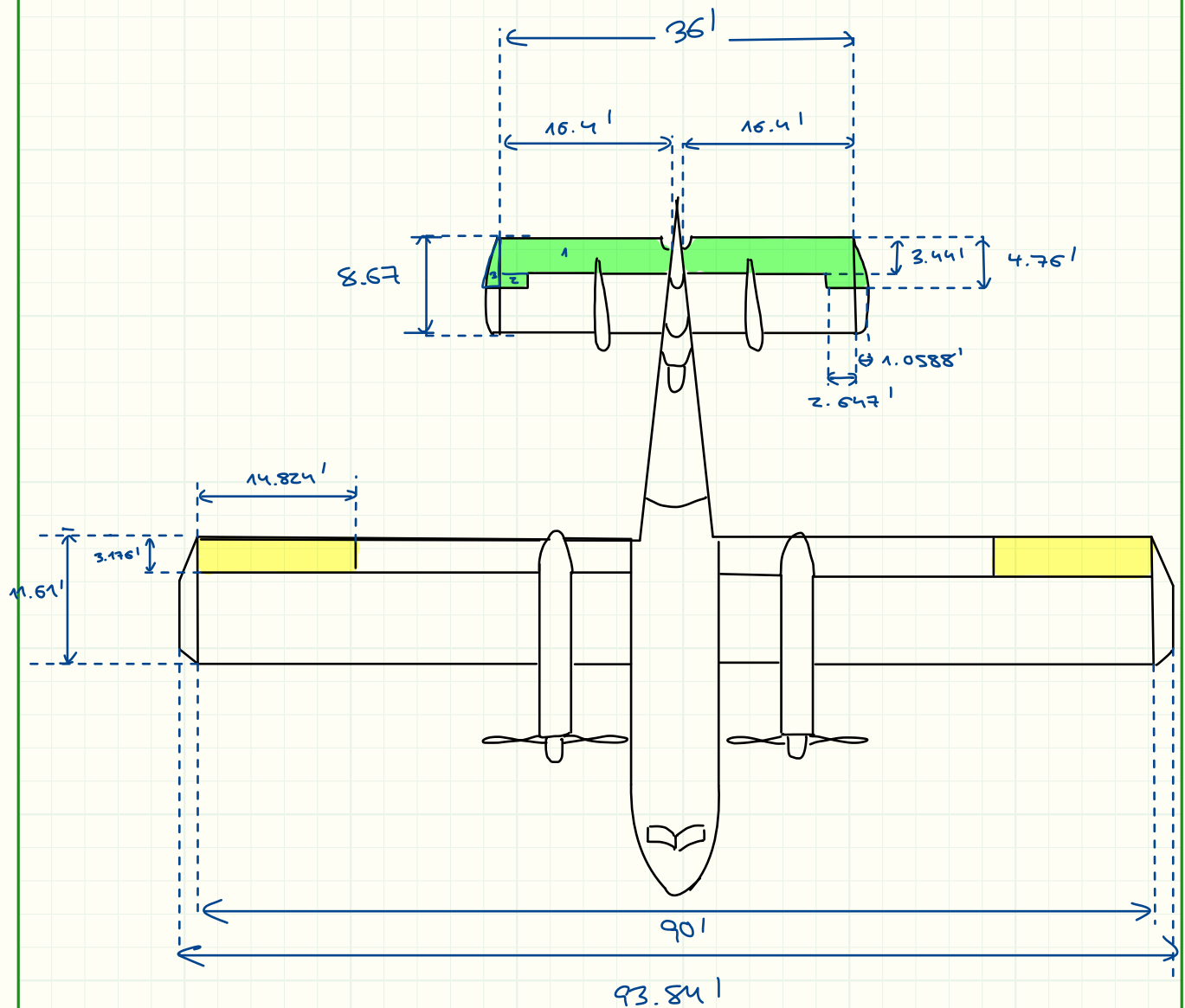


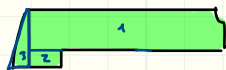




Aileron Surface Area : (rectangle)

$$S_a = (14.824)(3.176) = 47.0865 \text{ ft}^2$$

Elevator S.A



$$A_1 = (16.4)(4.76) = 78.172 \text{ ft}^2$$

$$A_2 = (4.76 - 3.44)(2.649) = 3.503 \text{ ft}^2$$

$$A_3 = \frac{1}{2} (1.0588)(4.76) = 2.522 \text{ ft}^2$$

$$S_e = 84.723 \text{ ft}^2$$

CONTROL SURFACES

Lateral Control — AILERONS

$$C_{l_{\delta a}} = \frac{2 C_{L_{\alpha, w}} \tau_a}{S_w b_w} \int_{y_i = 30.176}^{y_o = 45} \underbrace{C_a(y)}_{C_r = \text{rectangular}} y \, dy$$

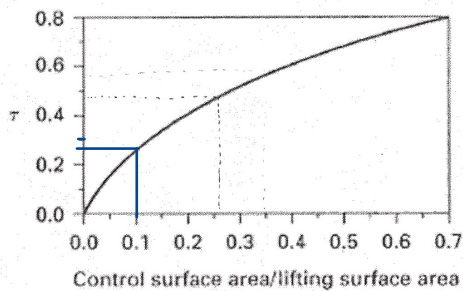


FIGURE 2.21
Flap effectiveness parameter.

$$\frac{C.S.A.}{L.S.A.} = \frac{S_a}{S_w} = \frac{2(47.08)}{1023.9} = 0.0919 \approx 0.1$$

$$\tau_a \approx 0.27$$

$$C_{l_{\delta a}} = \frac{2 (0.088 / \text{deg}) (0.27)}{(1023.9) (93.84)} \int_{30.176}^{45} 3.176 y \, dy$$

$$= 4.946 \times 10^{-4} \left[\frac{3.176 y^2}{2} \right]_{30.176}^{45} = 0.000876 / \text{deg}$$

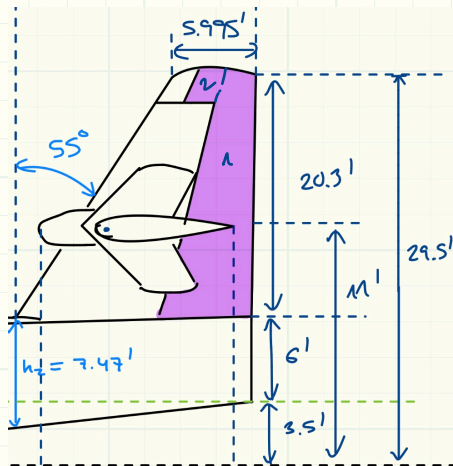
$$= 0.0502 / \text{rad}$$

Directional Control — Rudder

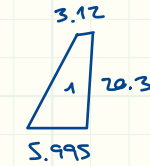
$$C_{n_{\delta r}} = -K \tau_{v,t} \bar{V}_2 C_{L_{\alpha, v,t}} \tau_z$$

$$\bar{V}_2 = \frac{S_v l_{v,t}}{S_w \bar{c}} = \frac{(221.567) (24.36)}{(1023.95) (11.61)} = 0.454$$

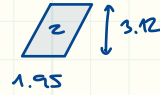
$$l_{v,t} = x_{a.c., vt} - x_{cg} = 50 - 25.64 = 24.36 \text{ ft}$$



Surface Area:



$$A_1 = \frac{(5.995 + 3.12) 20.3}{2} = 60.84 \text{ ft}^2$$



$$A_2 = (1.95)(3.12) = 6.084 \text{ ft}^2$$

$$S_r = 18.6 \text{ ft}^2$$

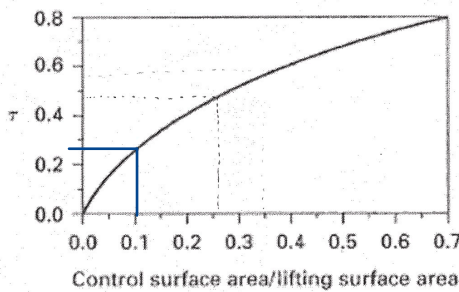


FIGURE 2.21
Flap effectiveness parameter.

$$\frac{C.S.A}{L.S.A} = \frac{S_r}{S_w} = \frac{18.6}{1023.9} = 0.018 \approx 0.02$$

$$\tau_r \approx 0.27$$

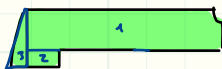
$$C_{n_{\delta_r}} = -K \tau_{v,t} \bar{V}_z C_{L_{\alpha,v,t}} \tau_z$$

$$C_{n_{\delta_r}} = -(0.88)(1)(0.454)(2.532/\text{rad})(0.27) = -0.273 \text{ rad}$$

$$= -0.00477/\text{deg}$$

Longitudinal Control — Elevator

Elevator S.A



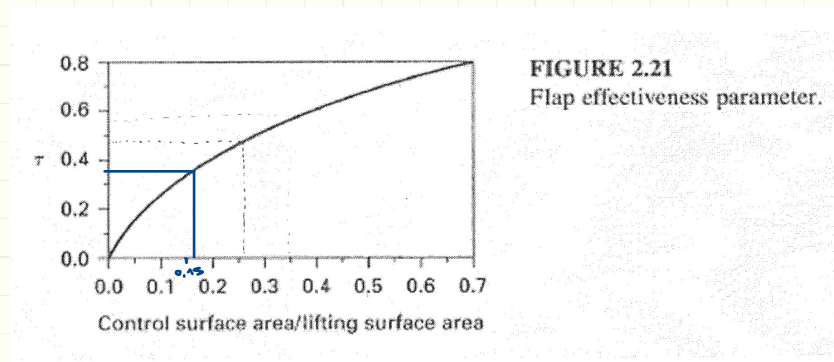
$$A_1 = (15.4)(4.76) = 73.304 \text{ ft}^2$$

$$A_2 = (4.76 - 3.44)(2.649) = 3.503 \text{ ft}^2$$

$$A_3 = \frac{1}{2} (1.0588)(4.76) = 2.522 \text{ ft}^2$$

$$S_e = 84.223 \text{ ft}^2$$

$$C_{m\delta_e} = - \bar{V}_{H,t} \eta C_{L\alpha,t} \tau_e$$



$$\frac{C.S.A}{L.S.A} = \frac{z(S_e)}{S_w} = \frac{z(84.2)}{1023.9} = 0.1645$$

$$\tau_e \approx 0.36$$

$$C_{m\delta_e} = - (0.80822)(1)(4.0127 \text{ /rad})(0.36) = \begin{matrix} -1.168 \text{ /rad} \\ -0.0204 \text{ /deg} \end{matrix}$$